

International Journal of Hydrogen Energy

CFD Analysis of Wall Function Effects on Flow Dynamics and Mixture Distribution in a Hydrogen Direct Injection Engine

--Manuscript Draft--

Manuscript Number:	
Article Type:	Full Length Article
Section/Category:	Heat Engines / Combustion / Heat Transfer
Keywords:	Hydrogen, Direct Injection, Computational Fluid Dynamics, Wall functions, Mixture Distribution
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Abstract:	Direct injection (DI) of gaseous hydrogen into the combustion chamber of internal combustion (IC) engines is considered a promising method to circumvent the issues of pre-ignition and backfire. However, a significant research focus is required to implement this technology in modern IC engines successfully. The present study focuses on a numerical investigation of mixture preparation in a spark-ignition IC engine where gaseous hydrogen is directly injected into the combustion chamber at a pressure of 100 bar. Numerical simulations on the engine were carried out at an engine speed of 1500 rev/min at a high tumble. The flow turbulence was modeled using the renormalized group (RNG) k-ϵ model. Different wall functions were used to model the flow and turbulence characteristics near the walls of the combustion chamber. The mixture distribution in the combustion chamber obtained by employing different wall functions was compared with the optical data acquired from planar laser-induced fluorescence (PLIF) measurements. The results showed that the Enhanced wall function and the Analytical wall function resulted in good agreement between the numerical analysis and optical measurement data compared to the other models.

To:

The Editor
International Journal of Hydrogen Energy

Dear Sir,

I am herewith submitting our paper titled **“CFD Analysis of Wall Function Effects on Flow Dynamics and Mixture Distribution in a Hydrogen Direct Injection Engine”** authored by Srinivasa Krishna Addepalli, Arunkumar V, and Sneha M S, for consideration for publication in your esteemed journal. The address of the corresponding author is as follows. This is an original paper that has neither previously, nor simultaneously, in whole or in part been submitted anywhere else.

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Thanking you,

With Regards,

Yours sincerely

Srinivasa Krishna Addepalli

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Highlights:

The impact of different wall functions on the mixture distribution is compared

The Enhanced and Analytical wall functions perform better than the Standard, Scalable, and Non-equilibrium wall functions

The Enhanced and Analytical wall functions result in higher velocity and higher mole fractions near the wall

CFD Analysis of Wall Function Effects on Flow Dynamics and Mixture Distribution in a Hydrogen Direct Injection Engine

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Abstract: Direct injection (DI) of gaseous hydrogen into the combustion chamber of internal combustion (IC) engines is considered a promising method to circumvent the issues of pre-ignition and backfire. However, a significant research focus is required to implement this technology in modern IC engines successfully. The present study focuses on a numerical investigation of mixture preparation in a spark-ignition IC engine where gaseous hydrogen is directly injected into the combustion chamber at a pressure of 100 bar. Numerical simulations on the engine were carried out at an engine speed of 1500 rev/min at a high tumble. The flow turbulence was modeled using the renormalized group (RNG) k - ϵ model. Different wall functions were used to model the flow and turbulence characteristics near the walls of the combustion chamber. The mixture distribution in the combustion chamber obtained by employing different wall functions was compared with the optical data acquired from planar laser-induced fluorescence (PLIF) measurements. The results showed that the Enhanced wall function and the Analytical wall function resulted in good agreement between the numerical analysis and optical measurement data compared to the other models.

Keywords: *Hydrogen, Direct Injection, Computational Fluid Dynamics, Wall functions, Mixture Distribution*

Nomenclature:

AMR	Adaptive Mesh Refinement
CAD	Crank Angle Degree
CFD	Computational Fluid Dynamics
CI	Compression-Ignition
DI	Direct Injection
ECN	Engine Combustion Network
IC	Internal Combustion
IVC	Intake Valve Closing
LES	Large Eddy Simulation
PFI	Port Fuel Injection